ECON 7201
Applied Econometrics
Assignment 2

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Due Date

November 5, 2026 at 11:59 PM

Directions

Answer all questions. Upload both a PDF and Quarto file to your website in a new blog post.

Git and GitHub

1. (a) Create a new R project in your **econ_3201** directory called **assignment_2**.
   (b) Download the assignment PDF and Quarto file the **assignment_2** folder.
   (c) Commit and push the changes to your **econ_3201** repository on GitHub.com.

LaTeX

Matrices are created in LaTeX using the `\begin{bmatrix}...\end{bmatrix}` command. To separate entries along the same row, use `&`. To end a line, use `\\`. To make vertical elipses (:), use `\vdots`. Practice writing the following matrices and vectors in LaTeX. Write the following matrices in LaTeX.

2. (a)

> Tip
>
> $$X^\top X = \begin{bmatrix} n & \sum_{i=1}^n x_{1i} & \sum_{i=1}^n x_{2i} \\ \sum_{i=1}^n x_{1i} & \sum_{i=1}^n x_{1i}^2 & \sum_{i=1}^n x_{1i}x_{2i} \\ \sum_{i=1}^n x_{2i} & \sum_{i=1}^n x_{1i}x_{2i} & \sum_{i=1}^n x_{2i}^2 \end{bmatrix}$$

(b)

Tip

$$\Omega = \begin{bmatrix} \sigma_1^2 & 0 & 0 & 0 \\ 0 & \sigma_2^2 & 0 & 0 \\ 0 & 0 & \sigma_3^2 & 0 \\ 0 & 0 & 0 & \sigma_4^2 \end{bmatrix}$$

R

3. In this question we compare standard errors based on (incorrect) asymptotic assumptions with those based on alternate (appropriate) estimator (White). Consider one sample drawn from the following data generating process (DGP) which we will simulate in `R`:

```
set.seed(123)
n <- 25
x <- rnorm(n,mean=0.0,sd=1.0)
beta0 <- 1
beta1 <- 0
## x is irrelevant in this model, the data generating process is as follows:
dgp <- beta0 + beta1*x
## The residual is heteroskedastic by construction
e <- x^2*rnorm(n,mean=0.0,sd=1.0)
y <- dgp + e
```

(a) Compute the OLS estimator of β_2 and its standard error using the `lm()` command in `R` for the model $y_i = \beta_1 + \beta_2 x_i + \epsilon_i$ based on the DGP given above.

Tip

I use `lm()` to estimate the model.

```
model <- lm(y ~ x)

round(
  coef(summary(model))["x", c("Estimate", "Std. Error")],
  4
)
```

```
  Estimate Std. Error
    0.4610     0.2563
```

The slope estimate is 0.4610. Its standard error is 0.2563.

(b) Next, compute the standard error of $\hat{\beta}_2$ by computing $\hat{\sigma}^2(X'X)^{-1}$ in R using matrix commands, and verify that the two standard error estimates are identical.

Tip

I calculate the covariance matrix using matrix operations.

```
X <- cbind(1, x)

beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
residuals_hat <- y - X %*% beta_hat

sigma2_hat <- as.numeric(
  t(residuals_hat) %*% residuals_hat
) / (n - ncol(X))

V_hat <- sigma2_hat * solve(t(X) %*% X)

se_matrix <- sqrt(V_hat[2, 2])
se_lm <- coef(summary(model))["x", "Std. Error"]

round(c(Matrix = se_matrix, lm = se_lm), 4)
```

```
Matrix     lm
0.2563 0.2563
```

```
isTRUE(all.equal(unname(se_matrix), unname(se_lm)))
```

```
[1] TRUE
```

Both methods give a standard error of 0.2563. The check returns TRUE.

(c) Compute White's heteroskedasticity consistent covariance matrix estimator using matrices in R and report the White estimator of the standard error of $\hat{\beta}_2$. Compare this with that from 3 (a) above.

Tip

I use the squared residuals to calculate White's covariance matrix.

```
XtX_inv <- solve(t(X) %*% X)
Omega_hat <- diag(as.numeric(residuals_hat)^2)

V_white <- XtX_inv %*% t(X) %*% Omega_hat %*%
  X %*% XtX_inv

round(V_white, 4)
```

```
                x
   0.0499 -0.0288
x -0.0288  0.2051
```

```
se_white <- sqrt(V_white[2, 2])

round(c(OLS = se_lm, White = se_white), 4)
```

```
   OLS  White
0.2563 0.4529
```

The White standard error is 0.4529. It is larger than the ordinary standard error of 0.2563. The errors are heteroskedastic, so the White estimator is appropriate.

4. Let $\hat{\theta}$ be an estimator for the population parameter θ. $\hat{\theta}$ is said to be unbiased if $E(\hat{\theta}) = \theta$. That is, if the mean of the sampling distribution of $\hat{\theta}$ is equal to the true population value.

   Consider the model

   $$y_i = \beta_0 + \beta_1 x_{1,i} - \beta_2 x_{2,i} + \epsilon_i.$$

   Lets provide empirical evidence that the ordinary least squares estimators $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ are unbiased estimators of β_0, β_1, β_2, respectively, using R.

   (a) Set the seed to 1, i.e., `set.seed(1)`.
   (b) Set the number of observations $n = 100$
   (c) Generate the following model

   $$y_i = 2 + 3.5 x_{1,i} - 9.2 x_{2,i} + \epsilon_i,$$

where $x_1 \sim N(3, 6)$, $x_2 \sim N(2, 4)$, and $\epsilon \sim N(0, 100)$. To create a normally distributed variable, use the `rnorm(n, mean, sd)` command in R.

(d) Estimate the model coefficients using the `lm()` command. (Search `?lm()` in the console for more info).

(e) Using a `for()` loop, replicate the model above $M = 1000$ times and save the coefficient estimates from each iteration.

(f) Using `hist()`, plot the sampling distributions of the coefficient estimates, β_1 and β_2.

(g) Add a vertical line to each figure at the mean of the respective variable. Search `?abline()` in your console.

(a)–(c)

I set the seed and generate 100 observations. I use the square roots of the variances as standard deviations.

```
set.seed(1)
n <- 100

x1 <- rnorm(n, mean = 3, sd = sqrt(6))
x2 <- rnorm(n, mean = 2, sd = sqrt(4))
epsilon <- rnorm(n, mean = 0, sd = sqrt(100))

y <- 2 + 3.5 * x1 - 9.2 * x2 + epsilon
```

(d)

I estimate the model using `lm()`.

```
fit <- lm(y ~ x1 + x2)
round(coef(fit), 4)
```

```
(Intercept)          x1          x2
     2.5297      3.5862     -9.4673
```

The estimates are close to the true coefficients. The coefficient on x_2 is negative. In the question, this coefficient is $-\beta_2$.

(e)

I generate a new sample in each iteration. I repeat this 1,000 times and save the estimates. I reverse the sign of the coefficient on x_2 to estimate β_2.

```r
M <- 1000

estimates <- matrix(NA_real_, nrow = M, ncol = 3)
colnames(estimates) <- c("beta0", "beta1", "beta2")

for (m in seq_len(M)) {
  x1 <- rnorm(n, mean = 3, sd = sqrt(6))
  x2 <- rnorm(n, mean = 2, sd = sqrt(4))
  epsilon <- rnorm(n, mean = 0, sd = sqrt(100))

  y <- 2 + 3.5 * x1 - 9.2 * x2 + epsilon

  fit_m <- lm(y ~ x1 + x2)
  coefficients <- coef(fit_m)

  estimates[m, ] <- c(
    coefficients[1],
    coefficients[2],
    -coefficients[3]
  )
}

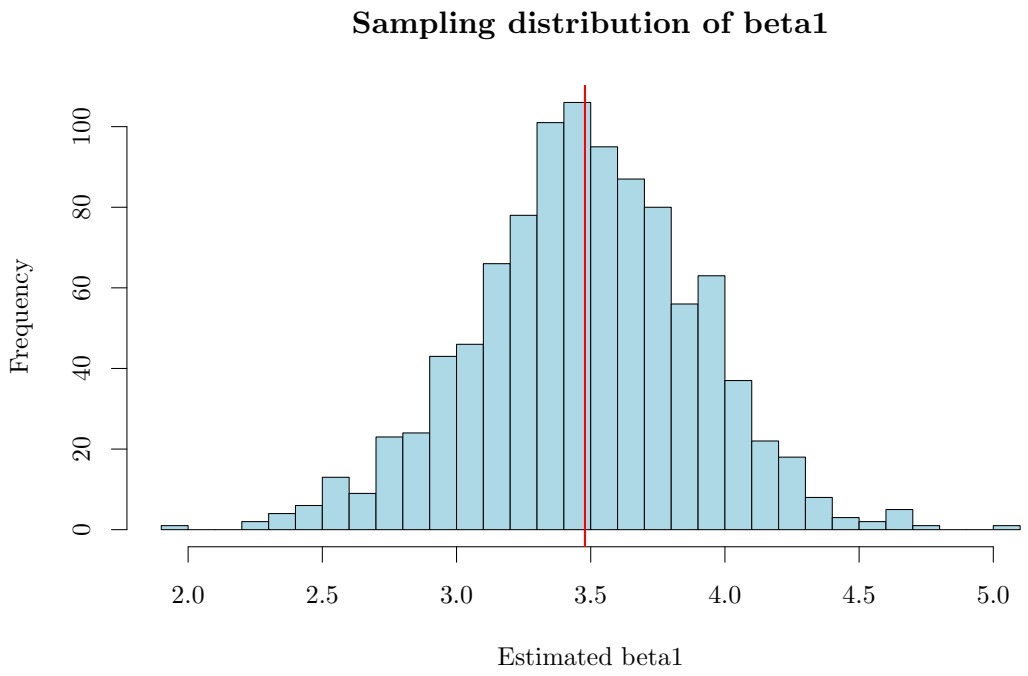
round(colMeans(estimates), 4)
```

```
 beta0  beta1  beta2
2.0075 3.4786 9.1861
```

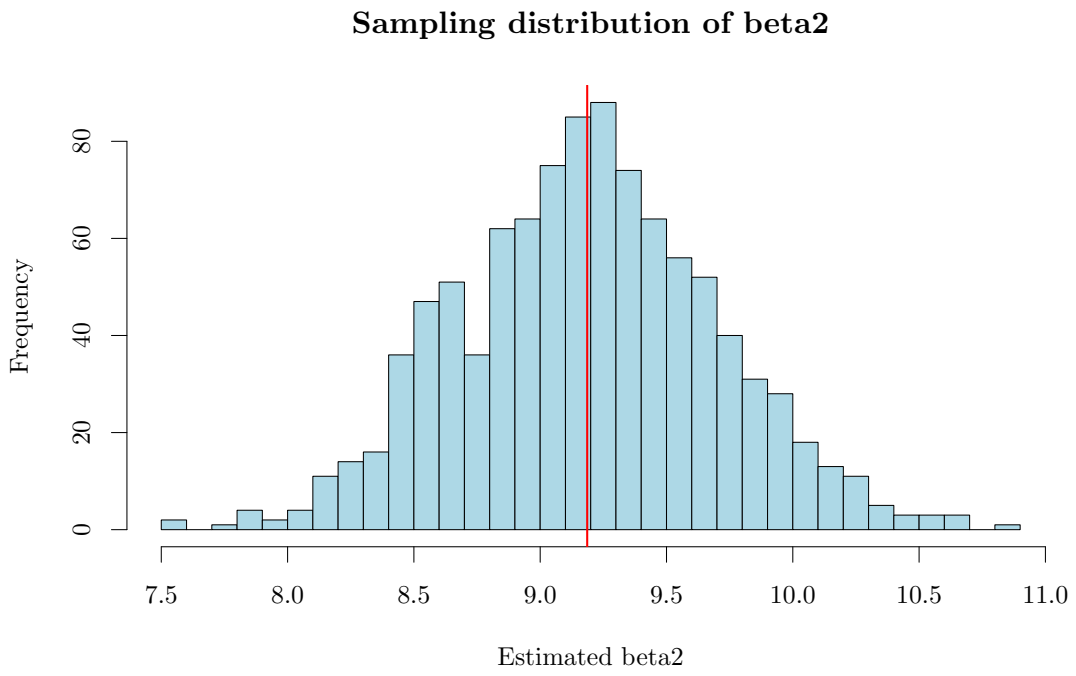
(f)–(g)

I plot the estimates of β_1 and β_2. The red lines show their simulation means.

```r
hist(
  estimates[, "beta1"],
  breaks = 30,
  main = "Sampling distribution of beta1",
  xlab = "Estimated beta1",
  col = "lightblue"
)
abline(
  v = mean(estimates[, "beta1"]),
  col = "red",
  lwd = 2
)
```



```
hist(
  estimates[, "beta2"],
  breaks = 30,
  main = "Sampling distribution of beta2",
  xlab = "Estimated beta2",
  col = "lightblue"
)
abline(
  v = mean(estimates[, "beta2"]),
  col = "red",
  lwd = 2
)
```



The simulation means are close to 2, 3.5, and 9.2. The histograms are centered near the true values. These results support unbiasedness, but a simulation alone does not prove it.